No. of Functional Features Included in the Solution

Date	04 JULY 2026
Team ID	
Project Name	OptiCrop: Smart Agricultural Production Optimization Engine
Maximum Marks	5 Marks

Functional Features Overview:

This document lists all the functional features included in the OptiCrop project solution. Each feature is described with its purpose, implementation status, and contribution to the overall solution.

S.No	Feature Name	Feature Description	Module / Component	Status	Marks Contrib.
1	User Registration & Login	Allows users to securely register, log in, and access the crop recommendation system.	Flask + SQLite	Done	High
2	Crop Recommendation	Predicts the most suitable crop based on Nitrogen, Phosphorus, Potassium, Temperature, Humidity, pH, and Rainfall using a trained Random Forest model.	ML Model	Done	High
3	Data Preprocessing	Removes duplicate records, handles missing values, performs feature selection, and prepares the dataset for model training.	Jupyter Notebook	Done	High
4	Exploratory Data Analysis	Performs Univariate, Bivariate, and Multivariate analysis using graphs for better understanding of the dataset.	Jupyter Notebook	Done	High
5	Machine Learning Model Training	Trains the Random Forest Classifier and evaluates prediction accuracy before deployment.	ML Model	Done	High
6	Model Persistence	Saves the trained model as **crop_model.pkl** using	Python	Done	High

		Pickle for fast prediction without retraining.			
7	Flask Web Application	Provides a responsive web interface for entering agricultural parameters and displaying crop prediction results.	Flask	Done	High
8	Database Management	Stores user registration and login details securely using SQLite.	SQLite Database	Done	Medium
9	Real-Time Prediction	Accepts user input through the web form and instantly predicts the recommended crop.	Flask + ML	Done	High

Feature Summary:

Metric	Count / Value
Total Features Planned	9
Total Features Implemented	9
Core / Must-Have Features	6 (Login, Crop Recommendation, Data Preprocessing, ML Model Training, Flask Application, Real-Time Prediction)
Additional / Nice-to-Have Features	3 (Database Management, Exploratory Data Analysis, Model Persistence)
Features Tested & Verified	9 (All features tested locally and verified successfully)

Feature Category Breakdown:

S.No	Category	Features in Category	Example Features
1	User Interface (UI)	3	Home Page, Login Page, Crop Prediction Page
2	Backend / Logic	4	Flask Routes, Prediction Engine, Input Validation, Session Management
3	Database / Storage	2	SQLite Database, User Authentication
4	Machine Learning / AI	3	Random Forest Model, Crop Recommendation, Real-Time Prediction
5	Data Analytics	2	Data Preprocessing, Exploratory Data Analysis
6	Security / Authentication	2	Password Hashing, User Login Authentication